Worksheet 8-7 & 8-8 A
Volumes of Solids with Cross Sections #1
AP Calculus Name_____

These questions are NOT calculator active questions. However, you may use DESMOS to graph the original region and may use a calculator for basic operations. Give the exact answer for each question.

1.) A solid S is built in such a way that its base is bounded by a circle of radius 3 meters and center at the origin. If each plane section perpendicular to a given diameter of the base is a square, find the volume of solid S. (Answer: 144 meters^3)

2.) Find the volume of a solid S if its base is bounded by the ellipse $x^2 + 4y^2 = 4$ and the cross sections perpendicular to the x-axis are squares. (Answer: $\frac{32}{3}$)

3.) Find the volume of a solid S if its base is bounded by the circle $x^2 + y^2 = 1$ and the cross sections perpendicular to the x-axis are equilateral triangles. (Answer: $\frac{4\sqrt{3}}{3}$)

4.) The base of a solid is the region bounded by the parabola $x^2 = 8y$ and the line $y = 4$, and each plane section perpendicular to the y-axis is an equilateral triangle. Find the volume of the solid. (Answer: $64\sqrt{3}$)

5.) Find the volume of a solid S if its base is bounded by the circle $x^2 + y^2 = 4$ and the cross sections perpendicular to the x-axis are semicircles. (Answer: $\frac{16\pi}{3}$)

6.) Find the volume of a solid S if its base is bounded by the circle $x^2 + y^2 = 16$ and the cross sections perpendicular to the x-axis are isosceles right triangles having the hypotenuse in the plane of the base. (Answer: $\frac{256}{3}$)

7.) Let region R be the area enclosed by $y = x^2 - 2$, the horizontal line $y = -4$, and the vertical lines $x = 0$ and $x = 4$. If the region R is the base of a solid, such that each cross section perpendicular to the x-axis is a semi-circle with a diameter on the base, find the volume of the solid. (Answer: $\frac{574\pi}{15}$)

8.) Let region R be the area enclosed by $y = x^2 - 2$, the horizontal line $y = -4$, and the vertical lines $x = 0$ and $x = 4$. If the region R is the base of a solid, such that each cross section perpendicular to the x-axis is a quarter-circle, find the volume of the solid. (Answer: $\frac{11}{15}$)

9.) Find the volume of the solid whose base is the semi-circle $y = \sqrt{16 - x^2}$ and whose cross sections perpendicular to the x-axis are squares with a side on the base. (Answer: $\frac{256}{3}$)

10.) Find the volume of the solid whose base is the region between $y = x^2$ and $y = 4$ and whose cross sections perpendicular to the x-axis are isosceles right triangles with the hypotenuse on the base. (Answer: $\frac{128}{15}$)